

LAB SAFETY

SCENARIO ANALYSIS

MAKE THE SAFE CALL

WHAT STUDENTS DO

Analyze 10 original laboratory situations, identify evidence of risk, choose the safest first action, and justify decisions using a consistent safety framework.

GRADES 9–11	CORE 45–55 MIN	PRINT + EDITABLE
CORE + HONORS	EXIT TICKET	FULL KEY

Original content • Printer-friendly • Structured reasoning

Welcome, Teacher

Use this ready-to-teach lesson for direct instruction, collaborative practice, a sub plan, or a readiness check.

LEARNING TARGET

I can notice evidence, describe the risk, choose the safest immediate action, and explain why it reduces harm.

SUGGESTED 50-MINUTE FLOW

Time	Action
5 min	Launch: What makes an action the safest first move?
8 min	Model Scenario 0 using Notice Risk Act Explain.
20 min	Pairs complete Scenarios 1-6; teacher checks reasoning.
12 min	Independent Scenarios 7-10.
5 min	Exit ticket and debrief. Honors extension adds 15-25 min.

QUICK START / SUB PLAN

Assign the Student Tool, all scenario pages, and the exit ticket. Students read the four-step framework independently and complete in order. Review with the key next class; a substitute should not improvise site-specific safety procedures.

DIFFICULTY PATH

Support 1/3: Scenarios 1-6 + category bank (35-45 min) | Core 2/3: all 10 (45-55 min) | Honors 3/3: all 10 + extension (60-75 min)

COMMON MISCONCEPTIONS

Reporting only after cleanup; returning excess reagent to stock; treating a colorless liquid as safe; naming a rule without explaining how it reduces the stated risk.

IMPORTANT

This resource supports instruction; it does not replace district policy, teacher supervision, SDS information, or site-specific emergency procedures.

Make the Safe Call

Name: _____ Class: _____ Date: _____

NOTICE RISK ACT EXPLAIN**1 NOTICE**

Underline the detail that signals a problem.

2 RISK

State what could happen or who could be exposed.

3 ACT

Choose the safest immediate action. Stop first when continuing could increase harm.

4 EXPLAIN

Connect the action to the evidence. Avoid vague answers such as "be careful."

CATEGORY BANK

PPE • chemical handling • heat/flame • glassware • housekeeping • emergency response

Scenario 0 model: A clearly identified water spill from a wash bottle is visible on a busy walkway. Evidence: the floor is wet. Risk: someone could slip. Action: block traffic and follow the teacher-approved cleanup procedure. Why: the route is protected while the hazard is removed correctly.

Scenarios 1-4

Notice the evidence, state the risk, choose the safest first action, and explain why it works.

1. Goggles on Standby

Maya begins measuring a dilute solution while her splash goggles rest on her forehead. Her partner is still collecting materials.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

2. Blocked Aisle

Two backpacks and a charging cable cross the aisle between a lab table and the exit.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

3. Mystery Beaker

An unlabeled beaker containing a clear liquid is left beside the balance. No one in the group knows who placed it there.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

4. Heating Direction

Eli warms a test tube while its opening points toward the student across the table.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

Scenarios 5-7

Notice the evidence, state the risk, choose the safest first action, and explain why it works.

5. Broken Glass

A cracked graduated cylinder breaks in the sink. A student reaches toward the pieces with a bare hand.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

6. Extra Reagent

After pouring too much reagent, a student prepares to return the excess to the stock bottle.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

7. Hair Near Heat

During burner setup, long loose hair hangs close to the work area and one sleeve is not secured.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

Scenarios 8–10

Notice the evidence, state the risk, choose the safest first action, and explain why it works.

8. Direct Smell

A student lifts an open container toward the nose to identify its odor without teacher direction.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

9. Splash Report

A small splash reaches a student's forearm. The student wipes it with a paper towel and says there is no need to interrupt class.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

10. Hot Plate Left On

A group finishes early and leaves the station while the hot plate remains on beside scattered papers.

Evidence noticed: _____

Risk / possible harm: _____

Safest first action: _____

Why this action reduces risk: _____

Prioritize and defend

Complete after all ten scenarios.

A. RISK RANKING

Choose the three scenarios that require the fastest intervention. Rank them 1-3. For each, explain severity, likelihood, and how quickly harm could occur.

Rank 1: Scenario ____ Reasoning:

Rank 2: Scenario ____ Reasoning:

Rank 3: Scenario ____ Reasoning:

B. CLAIM-EVIDENCE-REASONING

Claim: Which single habit would prevent the widest range of incidents in this set? Evidence: cite at least two scenarios. Reasoning: explain why the habit transfers across situations.

Ready for the lab?

Name: _____ Class: _____

1. You notice an unlabeled container at a shared station. What is the safest first action, and why?

2. Why is “be more careful” weaker than a specific safer-action statement?

3. Circle the category that needs more practice: PPE / chemicals / heat / glassware / housekeeping / emergencies

4. Write one question you still have before beginning laboratory work.

SELF-CHECK

I identified evidence [] I stated the risk [] I named a specific action [] I explained why []

Scenarios 1-4

Reasonable equivalent wording is acceptable when it follows local policy and does not add risk.

1. Goggles on Standby

Evidence: Chemical measuring has begun while Maya's goggles are on her forehead.

Risk: A splash could reach unprotected eyes.

Action: Stop measuring and wear splash goggles correctly before continuing.

Why: The eye barrier must be in place before exposure can occur.

2. Blocked Aisle

Evidence: Backpacks and a cable cross a travel route leading toward the exit.

Risk: Someone could trip, and movement during an emergency could be delayed.

Action: Stop traffic through the area and move the items to the assigned storage location.

Why: Removing the obstruction eliminates the trip hazard and keeps the exit route accessible.

3. Mystery Beaker

Evidence: The beaker has no label, and no one can identify the clear liquid.

Risk: Touching or testing an unknown substance could cause exposure or contamination.

Action: Do not touch or test it; keep others away and notify the teacher immediately.

Why: Avoiding contact prevents additional exposure while the teacher secures and identifies the material by site procedure.

4. Heating Direction

Evidence: The opening of a heated test tube points toward another person.

Risk: Hot liquid or vapor could eject toward that person.

Action: Stop heating; use the teacher-approved holder and point the opening away from all people.

Why: Stopping removes the immediate energy input, and safe orientation reduces direct exposure if material ejects.

Scenarios 5-7

Reasonable equivalent wording is acceptable when it follows local policy and does not add risk.

5. Broken Glass

Evidence: Broken glass is in the sink, and a bare hand is reaching toward it.

Risk: Sharp glass could cut the student.

Action: Do not pick it up by hand; alert the teacher and follow the designated broken-glass procedure.

Why: The designated tools and container prevent direct hand contact and isolate sharp waste.

6. Extra Reagent

Evidence: Previously poured reagent is about to be returned to the stock container.

Risk: The stock chemical could be contaminated, affecting later safety and results.

Action: Do not return it; ask the teacher for the correct waste procedure.

Why: Keeping used material out of the stock bottle protects the shared supply from contamination.

7. Hair Near Heat

Evidence: Loose hair and clothing are near a burner setup.

Risk: Hair or clothing could contact heat or flame and ignite or cause a burn.

Action: Turn off or do not light the heat source; secure hair and loose clothing first.

Why: Removing the ignition exposure before heating prevents contact with flame or hot equipment.

Scenarios 8–10 + extension

Reasonable equivalent wording is acceptable when it follows local policy and does not add risk.

8. Direct Smell

Evidence: The open container is being raised directly toward the student's nose without instruction.

Risk: A concentrated vapor could be inhaled.

Action: Move away, close the container if safe and directed, and ask the teacher; only waft when specifically instructed.

Why: Increasing distance and following teacher direction limits unnecessary inhalation exposure.

9. Splash Report

Evidence: A chemical reached skin, but the student only wiped it and did not report the exposure.

Risk: Chemical may remain on the skin, and delaying the correct response could increase injury.

Action: Alert the teacher immediately while beginning the room's designated skin-exposure rinse, unless the posted site/SDS procedure specifies otherwise.

Why: Rapid rinsing limits contact time while immediate reporting activates the correct site-specific response.

10. Hot Plate Left On

Evidence: An energized hot plate is unattended beside paper.

Risk: Someone could be burned, or nearby combustible material could ignite.

Action: Remain at the station. If trained and it is safe, use the normal control to turn off the hot plate; keep others clear and alert the teacher if it cannot be shut down safely.

Why: Immediate safe shutdown removes the heat source, while guarding the area prevents contact or ignition during escalation.

CORE RUBRIC – 3 POINTS PER SCENARIO

1 point: evidence and risk are specific and connected. 1 point: first action is safe, observable, and immediate. 1 point: explanation shows how the action reduces the stated risk.

HONORS EXEMPLAR

One defensible ranking is 4 first, 9 second, and 10 third. Other rankings earn credit when severity, likelihood, and immediacy are supported with scenario evidence.

CER MODEL

Claim: Pausing to check the setup before acting prevents the widest range of incidents. Evidence: Scenario 1 begins chemical handling before goggles are in place, and Scenario 6 returns used reagent toward shared stock. Reasoning: a deliberate check catches both personal-protection and contamination risks before exposure spreads.

EXIT TICKET GUIDANCE

1. Do not touch or test; keep others away and notify the teacher. 2. Specific actions are observable and directly reduce the identified risk. Items 3–4 are formative.

Sources, rights, and use

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ORIGINALITY

All student scenarios, questions, explanations, and layout are original. External sources informed factual accuracy and instructional scope; no source questions or passages were copied or adapted.

- American Chemical Society. Guidelines for Chemical Laboratory Safety in Secondary Schools and RAMP resources. <https://www.acs.org/education/policies/middle-and-high-school-chemistry/safety.html>
- American Chemical Society. Safety Equipment Guidelines. <https://www.acs.org/education/policies/middle-and-high-school-chemistry/classroom-and-lab-facilities/safety-equipment.html>
- American Chemical Society. Student Laboratory Code of Conduct. <https://institute.acs.org/acs-center/lab-safety/education-training/high-school-labs/student-lab-code-of-conduct.html>
- OpenStax Chemistry 2e, Chapter 1, used only for scope and terminology benchmarking. Current English edition license is CC BY-NC-SA 4.0; no content was adapted for this commercial resource. <https://openstax.org/books/chemistry-2e/pages/preface>

TERMS OF USE

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QUALITY CHECK

Before student use, confirm that emergency procedures and disposal language match your school, room, and available safety equipment.